

Practical - 5

Practical Title: Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition: Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

A. Theory

1. Are correct input types and attributes used?

Answer: Yes, semantic HTML5 input types and attributes are correctly used.

2. Are validation errors displayed near the relevant fields?

Answer: Yes, error messages are placed directly below each corresponding input field for clear visibility.

3. Is password strength checked?

Answer: Yes, password strength is checked using regular expressions requiring at least 8 characters, an uppercase letter, a lowercase letter and a number.

4. Is the form accessible using keyboard and screen-reader-friendly labels?

Answer: Yes, the form uses explicit <label for="..."> associations and native elements for accessibility and screen-reader labels.

B. Implementation/ Output/Result

Component	HTML input type	Regular Expression	Description
Full name	type="text"	/^[A-Za-z\s]{3,50}\$/	Only alphabets and spaces allowed (3-50 characters)
Email address	type="email"	/^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[a-zA-Z]{2,}\$/	Email format verification
Mobile number	type="tel"	/^[6-9]\d{9}\$/	Validates 10 digit mobile number starting with 6-9

Password	type="password"	/^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)[A-Za-z\d@\$!%*?&]{8,}\$/	Min. 8 chars with 1 uppercase, 1 lowercase and 1 digit
Confirm Password	type="password"	Equality check(password === confirmPassword)	Ensures both password fields match
Course	<select>	Value selection check(field.value !== '')	Validates that a course is chosen from the list.
Year of study	<select>	Value selection check(field.value !== '')	Validates that year is chosen
Gender	type="radio"	Radio state check	Ensures male, female, of other is selected

C. Evidence of Work (Screenshots)

StudentHub Portal Registration

Registration Details

Full Name
xyz abc

Email Address
xyz.abc@example.com

Mobile Number
9876543210

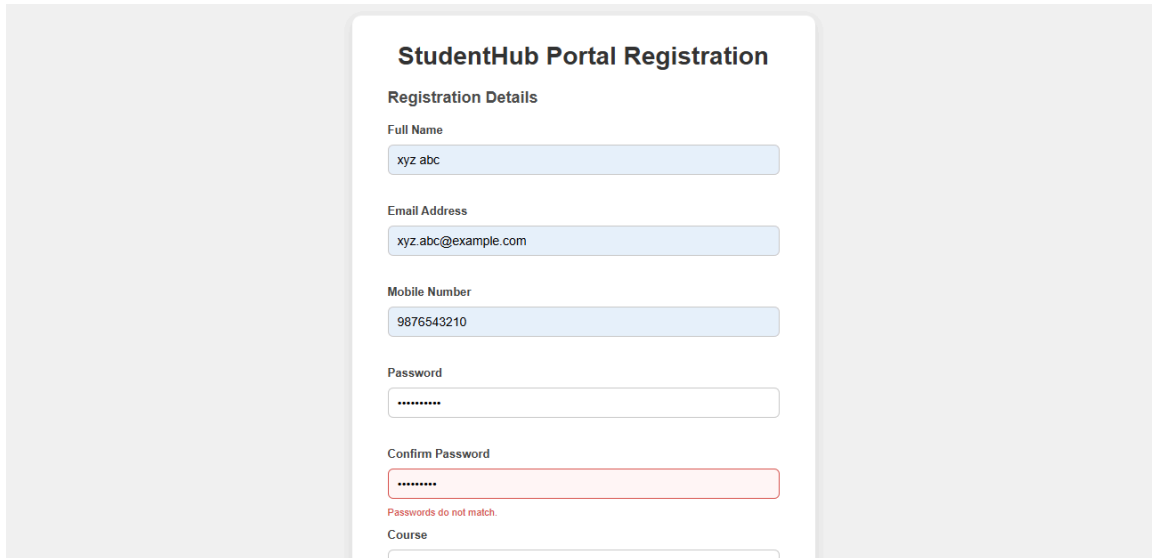
Password

Password must be at least 8 chars with 1 uppercase, 1 lowercase & 1 digit

Confirm Password

Course
Computer Engineering

Description: Figure 5.1 Password strength check in registration page



The image shows a web form titled "StudentHub Portal Registration". Under the heading "Registration Details", there are several input fields: "Full Name" (containing "xyz abc"), "Email Address" (containing "xyz.abc@example.com"), "Mobile Number" (containing "9876543210"), "Password" (containing "*****"), and "Confirm Password" (containing "*****"). A red error message "Passwords do not match." is displayed below the "Confirm Password" field. At the bottom, there is a "Course" field which is currently empty. The form is set against a light gray background.

Description: Figure 5.2 Confirm password check in registration page

D. GitHub Repository Link:

Link: <https://github.com/prep04-yl/StudentHub-Portal>

E. Challenges Faced:

Creating regex patterns to validate just what is needed and not be too restrictive or too simple.

F. Learning Outcome:

Learned form validation using regular expressions in Javascript and CSS styling for better user experience.

Faculty Remarks: (Keep this blank for faculty)